Systems Programmer

Zemlyanaya Daria

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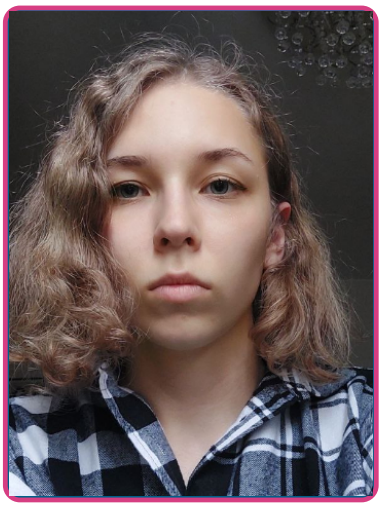

Passionate junior systems programmer with hands-on project experience. Developed multiple projects using C and x86-64 Assembly. Particularly interested in compiler internals, program performance optimization and virtual machine. Currently studying at MIPT.

Skills

Programming:
C, x86 Assembly, Python

Languages

Russian: Native
English: Intermediate (B1)

Education

Moscow Institute of Physics and Technology *Moscow, Russia* — 2024 - Present

- Faculty of Radioengineering and Cybernetics (FRKT)
- Specialization: System Programming
- Key courses: Computer Architecture, Operating Systems

Lyceum "Second School" *Moscow, Russia* — 2021 - 2024

- Physics and Mathematics specialization
- Multiple olympiad winner in mathematics and physics

School Achievements

- **Physics:**
  - Prize winner, "Step into the Future" Olympiad (2023, Grade 10)
- **Mathematics:**
  - Prize winner, "Financist" Olympiad (2024, Grade 11)
  - Prize winner, "Step into the Future" Olympiad (2024, Grade 11)
  - Prize winner, "Rosatom" Olympiad (2024, Grade 11)

Experience

Data Analytics Intern *MTS Digital* — 06/2024 - 08/2024

Big Data Internship

- Gained hands-on experience with PySpark for large-scale data processing
- Worked with Git version control in GitLab environment
- Assisted in data preparation and analysis for AI/ML projects
- Developed understanding of practical applications for AI models in business
- Participated in team meetings and project discussions

Tech Stack PySpark, Git/GitLab, Python, Big Data concepts

*Complete project portfolio available upon request. Key implementations linked in profile.

Key Course Projects @ Huawei/MIPT

Supervised by Ilya Dedinsky | 10+ system programming projects completed

CPU Emulator System *System Programming* **2024**

- **Pure C implementation**: Assembler compiler + stack-based VM
- **18 instructions**:
  - Arithmetic: ADD/SUB/MUL/DIV
  - Stack ops: PUSH/POP
  - Flow control: JMP/CALL/RET
- Features: label support, recursion examples (factorial)
- [Link to github]

Tech Stack C, Syntax Assembly

Low-Level printf Implementation *Assembly/C Integration* **2025**

- **C-ASM interface** via extern calls
- **Format support**: %d %s %c %% + numeric (%b/%o/%x)
- Unified number processing for base formats
- [Link to github]

Tech Stack Nasm, System V ABI

Mandelbrot Set Optimizer *SIMD Programming* **2025**

- **3 implementations**:
  - Scalar baseline
  - Auto-vectorized (-O3)
  - Manual AVX2 intrinsics (__m256)
- Interactive visualization with TxLib
- [Link to github]

Tech Stack C, AVX2 Intrinsics, TxLib